

Matthew Hong

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(650)-588-4858 • [LinkedIn](#) • [Engineering Portfolio](#)

Education

Northeastern University, Boston, MA - 3.76 GPA
Majoring in Mechanical Engineering with Minor in Business & EE

September 2025 - May 2029

London Scholar Program - First year studies in London

August 2025 – April 2026

Work Experience

Quanta Labs, Santa Clara, CA
Intern- Shock and Vibration Team

June 2023 - August 2023
May 2026 - August 2026

2023

- Wrote and organized Standard Operating Procedures to comply with company certifications
- Supported battery testing operations for automotive and energy sector clients
- Assisted with test setup, data collection, and reporting under the direction of senior engineers

2026

- Conducted shock and vibration testing and interfaced directly with aerospace, defense, and industrial clients to support project deliverables
- Designed and built a Flask/SQLite web application to manage a 125+ unit accelerometer inventory, replacing manual tracking with PIN-based checkout, calibration-due alerts, and full audit history
- Identified issues in test fixtures and procedures and proposed corrective solutions that were adopted by the team and resolved the underlying problems

ICode, Palo Alto, CA
Student Teacher

June 2024 - August 2024

- Designed and taught weekly Vex Robotics lessons for groups of 20 students aged 10-14
- Prepared and presented lesson plans to staff and parents, adapting content to varying skill levels
- Provided one-on-one instruction to reinforce engineering and problem-solving concepts

Engineering Projects

DREAMS Autonomous Catamaran USV
Co-Lead | Mechanical & Structural Engineer | Planning & External Communications

London, UK
December 2025 - April 2026

- Co-led mechanical and structural design for an autonomous catamaran USV integrating a Raspberry Pi and BME280 sensor suite for real-time atmospheric and water monitoring
- Modeled custom components in SolidWorks with bilateral tolerances for 3D-printed parts; designed a Van Veen grab sediment sampler with a complete engineering drawing package and BOM
- Achieved autonomous functionality during on-water testing at St Katharine Docks, London
- Coordinated with marina management to secure water access and led planning and external communications for a multidisciplinary team
- Continuing project scope through a collaboration with the Port of Oakland

Skills & Interests

Technical: Python, Java, AutoCAD, SolidWorks, Raspberry Pi, Flask, SQLite

Language: English (fluent), Spanish (conversational)

Laboratory: Shock/vibration testing equipment, 3D printers, laser cutters, lathe, hand tools

Interests: Rowing (4 years, Serra HS Crew Team), fly fishing, backpacking, F1, Golf